1 Appendix

1.1 Motivating The Posterior Density

To obtain a measure of uncertainty on our result, that is, to quantify how certain we are that the underlying probability $P(H_{any} \mid C, N)$ deviates from the independent-agent reference $P_{ind}(H_{any} \mid C, N)$, the following procedure is performed.

For a given experiment E, parametrised by C, N and O (where O denotes the other experimental hyperparameters that define the experimental run), we observe k trials involving at least one harmful action out of the n trials we run. That is, the empirical harmful-action probability for this experiment, E, is

$$\hat{p} = \frac{k}{n}. \tag{1}$$

Assuming that repeated trials under the same experimental configuration are independent and share the same underlying probability p, we can model the number of harmful-action outcomes using the binomial distribution. That is,

$$P(K = k \mid p, n) = L(p \mid k, n) = \binom{n}{k} p^k (1-p)^{n-k}. \tag{2}$$

However, we are not interested in the probability of observing k harmful-action outcomes given p. Instead, we are interested in what our observations tell us about the underlying harmful-action probability p^* for this experiment. Under the assumptions of our binomial model, the strong law of large numbers gives us that $\hat{p}$ converges almost surely to the underlying harmful-action probability, p^*, as the number of trials tends to infinity.

However, for finite n, uncertainty remains about the value of p^*. So, what we require is the posterior probability density,

$$\pi(p \mid k, n). \tag{3}$$

We use a probability density because p is a continuous parameter on the support $[0, 1]$. To find this posterior, we use Bayes' theorem. In general, for events A, B and conditioning information C, this is

$$P(A \mid B, C) = \frac{P(B \mid A, C) P(A \mid C)}{P(B \mid C)}. \tag{4}$$

For our continuous parameter p, the corresponding density form is

$$\pi(p \mid k, n) = \frac{P(K = k \mid p, n)\pi(p \mid n)}{P(K = k \mid n)} = \frac{L(p \mid k, n)\pi(p)}{P(K = k \mid n)}, \tag{5}$$

where we have assumed that $\pi(p \mid n) = \pi(p)$, since the number of trials n is fixed by our experimental design and provides no additional information about p.

The denominator is obtained using the continuous law of total probability:

$$P(K = k \mid n) = \int_0^1 P(K = k \mid p, n)\pi(p)\, dp. \tag{6}$$

It is the normalising constant for our posterior density, ensuring that

$$\int_0^1 \pi(p \mid k, n)\, dp = 1. \tag{7}$$

Therefore,

$$\pi(p \mid k, n) \propto L(p \mid k, n)\pi(p). \tag{8}$$

What is now required is for us to reason about what functional form would be sensible to choose for our prior probability density, $\pi(p)$.

1.2 Jeffreys' Prior

To specify this prior, we need to introduce desired properties that narrow down the space of possible functions. Consider the one-dimensional statistical manifold of binomial distributions,

$$\mathcal{M} = \{P_p : 0 < p < 1\}, \tag{9}$$

where P_p denotes the binomial distribution parametrised by p, with n fixed. Our prior assigns probability mass over this manifold. So what is a sensible way to define this statistical geometry? In other words, what is a sensible metric g_{pp} on $\mathcal{M}$? We write its line element as

$$ds^2 = g_{pp}dp^2. \tag{10}$$

One natural choice is to define infinitesimal separation on this manifold in terms of how statistically distinguishable neighbouring distributions are, with distributions that are easier to distinguish considered further apart.

We can motivate such a geometry using the KL divergence. Consider two distributions, P_p and P_q, with probabilities $P(K = k \mid p, n)$ and $P(K = k \mid q, n)$. The log-likelihood ratio between them, which measures how strongly a particular observation favours one distribution over the other, is

$$\Delta = \log \frac{P(K = k \mid p, n)}{P(K = k \mid q, n)}. \tag{11}$$

Here,

$$\Delta < 0, \qquad \Delta = 0, \qquad \Delta > 0, \tag{12}$$

tells us whether the observed data favour q, neither parameter, or p, respectively. We take the logarithm of the likelihood ratio because it converts products into sums, which is useful when combining evidence from independent observations. Since the logarithm is strictly increasing, it preserves which parameter is favoured.

Rather than relying on one realised count k, we can average over all possible outcomes $K = 0, \ldots, n$, weighted by their probabilities under P_p. This measures the expected log-likelihood advantage of p over q, assuming P_p generates the data. We therefore define the Kullback–Leibler (KL) divergence as,

$$D_{\mathrm{KL}}(P_p \| P_q) = \mathbb{E}_{K \sim P_p}[\Delta] = \sum_{k=0}^{n} P(K = k \mid p, n) \log \frac{P(K = k \mid p, n)}{P(K = k \mid q, n)}. \tag{13}$$

For our case, we can now substitute in our binomial distribution, (2). This gives us,

$$D_{\mathrm{KL}}(P_p \| P_q) = n \left[p \log \left(\frac{p}{q} \right) + (1 - p) \log \left(\frac{1 - p}{1 - q} \right) \right]. \tag{14}$$

If we then assume $q = p + dp$ and Taylor expand our result to second order in dp, we arrive at,

$$D_{\mathrm{KL}}(P_p \| P_{p+dp}) = \frac{1}{2} \frac{n}{p(1 - p)} (dp)^2 + O((dp)^3). \tag{15}$$

We define the Fisher–Rao metric using the quadratic term of the KL divergence, adopting the conventional definition

$$ds^2 \equiv 2 D_{\mathrm{KL}}^{(2)}, \tag{16}$$

where $D_{\mathrm{KL}}^{(2)}$ denotes only the second-order term of the Taylor expansion. Hence, for our binomial model,

$$ds^2 = \frac{n}{p(1 - p)} (dp)^2. \tag{17}$$

Comparing this with $ds^2 = g_{pp}(dp)^2$, we identify the metric component g_{pp} with the Fisher information $I_n(p)$:

$$g_{pp} = I_n(p) = \frac{n}{p(1-p)}. \tag{18}$$

This gives us a statistical geometry but not yet a defined prior. To do this, we need to introduce another criterion that narrows down the space of possible functions. The criterion we now specify is the Jeffreys choice to assign equal prior probability to equal intervals of Fisher–Rao length. That is, the prior probability mass assigned to an infinitesimal interval is proportional to its statistical length:

$$\pi_J(p)\, dp \propto ds. \tag{19}$$

For an increasing displacement $dp > 0$, the line element gives,

$$ds = \sqrt{g_{pp}}\, dp = \sqrt{\frac{n}{p(1-p)}}\, dp. \tag{20}$$

Therefore,

$$\pi_J(p) \propto \frac{1}{\sqrt{p(1-p)}}, \tag{21}$$

where the factor $\sqrt{n}$ is constant with respect to p and is absorbed into the normalisation.

Therefore,

$$\int_0^1 \pi_J(p)\, dp = A \int_0^1 p^{-1/2}(1-p)^{-1/2}\, dp. \tag{22}$$

Recognising that

$$\int_0^1 p^{-1/2}(1-p)^{-1/2}\, dp = B\left(\frac{1}{2}, \frac{1}{2}\right), \tag{23}$$

where B is the Beta function, defined by

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \tag{24}$$

Since $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(1) = 1$, we have

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = \pi. \tag{25}$$

Moreover, since our prior must satisfy

$$\int_0^1 \pi_J(p)\, dp = 1, \tag{26}$$

we arrive at $A = 1/\pi$, and our normalised Jeffreys prior is

$$\pi_J(p) = \frac{1}{\pi\sqrt{p(1-p)}}, \qquad 0 < p < 1. \tag{27}$$

This is precisely the probability density of the Beta$(1/2, 1/2)$ distribution. Finally, to arrive at our posterior distribution, we need to return to Bayes' theorem, that is,

$$\pi(p \mid k, n) \propto L(p \mid k, n)\pi_J(p) \propto p^{k-1/2}(1-p)^{n-k-1/2}. \tag{28}$$

Now, given that

$$\int_0^1 \pi(p \mid k, n)\, dp = 1, \tag{29}$$

this implies that,

$$\int_0^1 p^{k-1/2}(1-p)^{n-k-1/2}\, dp = \frac{1}{C_{\text{post}}}. \tag{30}$$

Recognising that we can again rewrite this in terms of a Beta function, it follows that

$$\frac{1}{C_{\text{post}}} = B\left(k + \frac{1}{2}, n - k + \frac{1}{2}\right). \tag{31}$$

Therefore,

$$\boxed{\pi(p \mid k, n) = \frac{p^{k-1/2}(1-p)^{n-k-1/2}}{B\left(k + \frac{1}{2}, n - k + \frac{1}{2}\right)}}. \tag{32}$$

1.3 Using the Posterior to Calculate Confidence

Recall that our aim is to quantify the uncertainty in the deviation between the underlying probability $P(H_{any} \mid \boldsymbol{C}, N)$ and the independent-agent reference $P_{ind}(H_{any} \mid \boldsymbol{C}, N)$. More precisely, we want to quantify the posterior probability that the deviation has a particular sign.

Let $\boldsymbol{C} = (C_1, \ldots, C_N)$ denote the self-conceptions of the N agents in a group, and let $\mathcal{S}$ denote the set of distinct self-conceptions within that group. For each $c \in \mathcal{S}$, let m_c denote the number of agents with self-conception c, such that $\sum_{c\in\mathcal{S}} m_c = N$.

Let p_g denote the underlying harmful-action probability in the communicating-agent experiment, and let p_c denote the underlying single-agent harmful-action probability for self-conception c. Assuming agents act independently in the reference model, the independent-agent reference is

$$r(\boldsymbol{p}) = 1 - \prod_{c \in \mathcal{S}} (1 - p_c)^{m_c}, \tag{33}$$

where $\boldsymbol{p} = (p_c)_{c \in \mathcal{S}}$ denotes the distinct single-agent probabilities.

For example, for two agents with self-conceptions A and B, the reference is

$$r_{AB} = 1 - (1 - p_A)(1 - p_B). \tag{34}$$

When both agents share the same self-conception, this reduces to $r_{AA} = 1 - (1 - p_A)^2$. More generally, for N agents sharing self-conception C, we recover $r = 1 - (1 - p_C)^N$.

The binomial independence assumption concerns repeated experimental trials, not the agents within a communicating group. Independence between agents is instead an assumption of the reference model. Interpreting the deviation as an interaction effect also requires the single-agent and communicating-agent experiments to be comparable in relevant respects, including the agent model, task distribution and sampling procedure.

We define the deviation as

$$\Delta_{\text{int}} = p_g - r(\boldsymbol{p}) = p_g - \left[1 - \prod_{c \in \mathcal{S}} (1 - p_c)^{m_c}\right]. \tag{35}$$

In other words, we want to find $P(\Delta_{\text{int}} > 0 \mid D, \boldsymbol{C}, N)$, where $D = (D_g, \{D_c\}_{c \in \mathcal{S}})$ denotes our observed data from the communicating-agent experiment and the relevant single-agent experiments.

We find this value by integrating the joint posterior density over the region where $p_g > r(\boldsymbol{p})$. Assuming independent priors and conditionally independent datasets, the joint posterior factorises, giving

$$\begin{aligned} &P(\Delta_{\text{int}} > 0 \mid D, \boldsymbol{C}, N) \\ &= \int_{[0,1]^{|\mathcal{S}|}} \int_{r(\boldsymbol{p})}^{1} \pi_g(p_g \mid D_g, \boldsymbol{C}, N) \\ &\qquad \times \prod_{c \in \mathcal{S}} \pi_c(p_c \mid D_c, c)\, dp_g \prod_{c \in \mathcal{S}} dp_c. \end{aligned} \tag{36}$$

Here, π_g denotes the posterior density of p_g for communicating agents, while π_c denotes the posterior density of p_c for a single agent with self-conception c. The independent-agent reference is determined by the single-agent probabilities and does not require a separate prior.

Both posterior densities follow from the Jeffreys prior derived above. Writing $D_g = (k_g, n_g)$ and $D_c = (k_c, n_c)$, we have

$$\pi_g(p_g \mid D_g, \boldsymbol{C}, N) = \frac{p_g^{k_g - 1/2}(1-p_g)^{n_g - k_g - 1/2}}{B\left(k_g + \frac{1}{2}, n_g - k_g + \frac{1}{2}\right)}, \tag{37}$$

and, for each $c \in \mathcal{S}$,

$$\pi_c(p_c \mid D_c, c) = \frac{p_c^{k_c - 1/2}(1-p_c)^{n_c - k_c - 1/2}}{B\left(k_c + \frac{1}{2}, n_c - k_c + \frac{1}{2}\right)}. \tag{38}$$

Substituting these into our expression for the probability of a positive deviation gives

$$\boxed{\begin{aligned} &P(\Delta_{\text{int}} > 0 \mid D, \boldsymbol{C}, N) \\ &= \int_{[0,1]^{|\mathcal{S}|}} \int_{r(\boldsymbol{p})}^{1} \frac{p_g^{k_g - 1/2}(1-p_g)^{n_g - k_g - 1/2}}{B\left(k_g + \frac{1}{2}, n_g - k_g + \frac{1}{2}\right)} \\ &\quad \times \prod_{c \in \mathcal{S}} \frac{p_c^{k_c - 1/2}(1-p_c)^{n_c - k_c - 1/2}}{B\left(k_c + \frac{1}{2}, n_c - k_c + \frac{1}{2}\right)} \, dp_g \prod_{c \in \mathcal{S}} dp_c. \end{aligned}} \tag{39}$$

This is the posterior probability that the communicating-agent harmful-action probability exceeds the independent-agent reference.

Since our posterior distributions are continuous, the probability of a negative deviation is

$$P(\Delta_{\text{int}} < 0 \mid D, \boldsymbol{C}, N) = 1 - P(\Delta_{\text{int}} > 0 \mid D, \boldsymbol{C}, N). \tag{40}$$

These probabilities quantify support for the direction of the deviation, conditional on our binomial model, Jeffreys priors, and independence assumptions. They do not quantify the probability of any nonzero deviation, which is one under this continuous model.

1.4 Computation Details

In practice, we can simplify the preceding integral by recognising that the inner integral over p_g is the complementary cumulative distribution function of a Beta distribution.

Writing

$$
\begin{aligned}
a_g &= k_g + \frac{1}{2}, \quad b_g = n_g - k_g + \frac{1}{2}, \\
a_c &= k_c + \frac{1}{2}, \quad b_c = n_c - k_c + \frac{1}{2},
\end{aligned}
\tag{41}
$$

for each distinct self-conception c, and defining the regularised incomplete Beta function as

$$
I_x(a,b) = \frac{1}{B(a,b)} \int_0^x t^{a-1}(1-t)^{b-1}\, dt, \tag{42}
$$

we can reduce our expression to

$$
\boxed{
\begin{aligned}
&P(\Delta_{\text{int}} > 0 \mid D, \boldsymbol{C}, N) \\
&= \int_{[0,1]^{|\mathcal{S}|}} \left[1 - I_{r(\boldsymbol{p})}(a_g, b_g)\right] \\
&\qquad \times \prod_{c \in \mathcal{S}} \frac{p_c^{a_c - 1}(1-p_c)^{b_c - 1}}{B(a_c, b_c)} \prod_{c \in \mathcal{S}} dp_c.
\end{aligned}
}
\tag{43}
$$

The remaining integral is evaluated numerically using quadrature, retaining uncertainty in all distinct underlying harmful-action probabilities. Its dimensionality is determined by the number of distinct self-conceptions in the group.

For example, when all N agents share self-conception C, the expression reduces to the one-dimensional integral

$$
\begin{aligned}
&P(\Delta_{\text{int}} > 0 \mid D, C, N) \\
&= \int_0^1 \left[1 - I_{1-(1-p_C)^N}(a_g, b_g)\right] \\
&\qquad \times \frac{p_C^{a_C - 1}(1-p_C)^{b_C - 1}}{B(a_C, b_C)}\, dp_C.
\end{aligned}
\tag{44}
$$

Alternatively, for two agents with distinct self-conceptions A and B, we obtain

$$
\begin{aligned}
&P(\Delta_{\text{int}} > 0 \mid D, A, B) \\
&= \int_0^1 \int_0^1 \left[1 - I_{1-(1-p_A)(1-p_B)}(a_g, b_g)\right] \\
&\qquad \times \frac{p_A^{a_A - 1}(1-p_A)^{b_A - 1}}{B(a_A, b_A)} \\
&\qquad \times \frac{p_B^{a_B - 1}(1-p_B)^{b_B - 1}}{B(a_B, b_B)}\, dp_A\, dp_B.
\end{aligned}
\tag{45}
$$

Importantly, repeated agents with the same self-conception share the same underlying probability parameter. Their posterior uncertainty is therefore represented once, rather than by independent draws for each agent.